Original Mixed-radix: Final Evidence and Closure

A real mechanism, but no stable natural-data method

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1. The Answer First

Final result for the tested method

The original fixed-adjacent mixed-radix method is **not competitive on SIFT1M or GIST1M** under the frozen Recall–speed evaluation.

- At 32 bytes per database vector, it becomes exactly the power-of-two control.
- At 64 bytes per database vector, Recall changes by at most about one tenth of one percentage point, with both positive and negative signs.
- A synthetic positive control now confirms that the intended mechanism itself is real and reaches the query result.

Interpretation

The implementation can exploit non-power-of-two cardinalities. The tested natural data simply does not contain enough of this structure for a useful win.

2. Scope Correction: Which "Attempt 4" Is This?

Original idea: this deck

Two adjacent scalar quantizers share one packed word. Their numbers of levels may be arbitrary positive integers.

Separate branch: not this result

The structured-2D method learns a reusable two-dimensional shape with affine transforms. It was evaluated separately and is currently set aside.

Why the distinction matters

Mixed-radix changes how a fixed number of bit patterns is divided between two scalar coordinates. It does **not** learn a shared 2D vector codebook.

3. The Original Intuition

Four bits give exactly $2^4 = 16$ possible labels for one coordinate pair.

Power-of-two allocation

Each coordinate receives 1, 2, 4, 8, ... levels. A typical balanced split is $4 \times 4 = 16$.

Mixed-radix allocation

Allow any integers K_1, K_2 as long as

$$K_1 K_2 \leq 16.$$

This includes $3 \times 5 = 15$.

Hypothesis

If one coordinate really needs 3 values and the other needs 5, mixed-radix can retain all 15 combinations. Separate binary rounding would need $4 \times 8 = 32$ states and cannot fit in four bits.

4. Running Example: Pack a 3×5 Grid

Let $z_1 \in \{0, 1, 2\}$ and $z_2 \in \{0, 1, 2, 3, 4\}$. Store

$$\text{label} = z_1 + 3z_2.$$

	$z_2 = 0$	1	2	3	4
$z_1 = 0$	0	3	6	9	12
$z_1 = 1$	1	4	7	10	13
$z_1 = 2$	2	5	8	11	14

Every real combination has a unique label; label 15 is simply unused.

What the power-of-two control must do

With only 16 labels it may choose 4×4, but then five distinct values of the second coordinate must share only four reconstructed values. At least two real prototypes can receive the same stored code.

5. Training, Encoding, and Querying

```
Training, using base/index data only:
  for each adjacent coordinate pair:
    estimate scalar reconstruction cost for each level count
    choose K1, K2 minimizing total cost, subject to K1*K2 <= 2^B

Encoding one database vector:
  z1 = nearest scalar centre on coordinate 1
  z2 = nearest scalar centre on coordinate 2
  store label = z1 + K1*z2

Querying:
  build the complete 2^B-entry distance table for each pair
  decode one label per pair, add table values, keep the best 100 IDs
```

Controlled difference

From here on, A means arbitrary integer level counts and D means the power-of-two control. They use the same pairs, bytes, candidates, table builder, and scan loop. Only the allowed values of K_1, K_2 differ.

6. What We Actually Compared

A: arbitrary	arbitrary integer level counts, such as $15 \times 17 = 255$
D: power-of-two	identical method, but each level count must be a power of two
PQ / OPQ	Product Quantization / Optimized Product Quantization baselines
coordinate view	first 128 PCA residual coordinates, forming 64 pairs
data	SIFT1M and GIST1M
payload	exactly 32 or 64 code bytes per database vector
index partitions	database divided into 1,024 or 4,096 coarse regions
quality	Recall@100: how many of the true nearest 100 are returned
speed	single-query latency and 12-core queries per second
search effort	nine fixed numbers of regions searched per query

Fairness rule

A and D search exactly the same candidates and perform the same table work. The allocation was fixed before reading query results.

7. What the Natural-data Allocator Chose

The 128-coordinate view forms 64 coordinate pairs; each pair contains two PCA coordinates and shares one packed label.

32-byte vectors: 4 bits per pair

All 64 coordinate pairs selected 4×4. Therefore A and D were literally the same representation and returned identical rankings.

64-byte vectors: 8 bits per pair

Depending on the dataset and index configuration, only 10–24 of the 64 coordinate pairs differed from 16×16. Almost all were 15×17 or 17×15, using 255 of 256 paid states.

Early mechanism diagnosis

The reconstruction-based allocator found no large unused-cardinality problem on SIFT/GIST. At 64 bytes it mostly moved only one level from one coordinate to its partner.

8. Natural-data Recall Result

At the same number of searched database regions, A-minus-D Recall@100 was:

Dataset	total index partitions	minimum	maximum
GIST1M	1,024	-0.00096	+0.00054
GIST1M	4,096	-0.00106	+0.00017
SIFT1M	1,024	-0.000097	+0.000066
SIFT1M	4,096	-0.000033	+0.000436

Meaning in plain language

The largest gain was 0.054 percentage points; the largest loss was 0.106 percentage points. The sign changed across datasets, index sizes, and search ranges, so there is no stable quality improvement.

9. Speed and Baseline Result

- A/D throughput ratio: 0.9918 to 1.0069; configuration medians are essentially 1.
- A/D p95 latency ratio: 0.9897 to 1.0251.
- This is expected: A and D deliberately execute the same complete-table query routine and nearly the same representation.
- PQ and OPQ generally reach higher Recall in the useful high-quality region.

Natural-data decision

The fixed-adjacent mixed-radix method provided neither clearly higher Recall at similar speed nor clearly higher speed at similar Recall. A tiny nonzero difference is not a database-systems contribution.

Complete matrix: 512 accepted passes; 32.36 CPU-hours; 17.09 wall-hours; all eight correctness tests pass.

10. Why Add a Synthetic Positive Control?

The natural result left two different explanations:

1. the mixed-radix mechanism is valid, but the required data structure is absent or weak; or
2. the implementation cannot transmit a radix advantage into query ranking, even when such an advantage exists.

Decisive construction

Create exact Cartesian prototypes with the cardinalities required by the intuition. Put 100 identical database vectors at every prototype, place one query exactly on each prototype, search every candidate, and ask whether all 100 true duplicates are recovered.

Positive cases are 3×5 in 4 bits and 15×17 in 8 bits. Matching 4×4 and 16×16 cases are zero-effect controls.

11. Synthetic Result: The Mechanism Reaches Recall

Case	A shape	D shape	merged prototype pairs A/D	Recall A/D
positive B4	3×5	4×4	0 / 3	**1.000 / 0.800**
null B4	4×4	4×4	0 / 0	1.000 / 1.000
positive B8	15×17	16×16	0 / 15	**1.000 / 0.941**
null B8	16×16	16×16	0 / 0	1.000 / 1.000

Validity checks

In both null controls A and D produced identical rankings. In both positive cases the rankings differed exactly where D merged prototypes. Two complete runs were byte-for-byte identical.

12. Put the Two Results Together

Question	Natural SIFT/GIST	Synthetic witness
Can A choose non-dyadic radices?	rarely, and only weakly at 64 bytes	yes, exactly 3×5 and 15×17
Does this avoid collisions?	no material measured benefit	yes, 3 and 15 control collisions removed
Does it improve Recall?	tiny, mixed-sign changes	yes, by 0.200 and 0.0588
Does it beat PQ/OPQ?	no better Recall–speed trade-off	not tested or claimed

Combined conclusion

The mechanism is real but its necessary structure is not sufficiently strong in the tested natural representation. The synthetic result explains the negative result; it does not reverse it.

13. Coordinate Regrouping Was Not the Missing Contribution

Allowing non-adjacent coordinate pairs improved Recall substantially over the fixed pairing $(0,1),(2,3),\ldots$. We then compared our minimum-cost perfect matching with the variance-based grouping rule from Optimized Product Quantization (OPQ).

	search 64 cells	search 1,024 cells
SIFT1M: matching minus OPQ grouping	+0.000253	+0.000349
GIST1M: matching minus OPQ grouping	+0.000320	-0.000160

Interpretation

"Do not pair coordinates mechanically" is useful, but already explained by known transform-coding practice. Arbitrary integer radices still add at most about 0.00057 Recall over the identical power-of-two pairing.

14. Why Reconstruction Error Was Not Enough

Recall depends on the ordering of complete estimated distances, not on the average reconstruction error of one coordinate block. Here a segment means a block of coordinates encoded and scored together.

The diagnostic question

For each database vector being re-encoded, replace one block's approximate distance by its true distance while leaving every other block approximate. How many wrongly ordered vector pairs become correct?

- This is an **idealized exact-distance control**: it uses information that a deployable compressed code does not store.
- A single coordinate block repaired only about 43–44% of ordering errors and missed the frozen absolute-improvement rule.
- Jointly replacing a 192-coordinate block at 6 bits per coordinate and a 320-coordinate block at 4 bits per coordinate repaired **71.1–75.6%**.

What this did and did not establish

The two blocks jointly contain many potentially removable ordering errors. True-distance replacement is not an encoder; the Recall ordering test can make separate errors look jointly important when they cancel each other.

15. Can Base Data Learn a Joint Encoder?

With the existing SAQ index layout fixed, every legal code for the first block can be combined with every legal code for the second block. Standard reconstruction and worst-case errors still split into one objective per block; joint benefit can only come from error correlations in chosen test directions.

Illustrative cancellation

On training directions, two block errors might be (+2) and (-2), so the total error is zero. On different directions they might become (+2) and (+2), so the same jointly selected pair has total error four. A cancellation seen during training need not survive new directions.

Falsifiable hypothesis

Choose the two codes jointly using directions constructed from base vectors, not benchmark queries. Freeze the choices and test them on a disjoint half. Require at least 5% lower mean squared estimator error than separate selection in both train/test swaps, and improvement on at least 60% of database vectors.

16. Train on One Half, Test on the Other, Then Swap

No benchmark query or ground truth was read. Each listed database vector is one whose two stored block codes are being reconsidered.

```
for each of 32 fixed database vectors:
  alternatives = stored code + every legal one-coordinate +/-1 move
  train on base-derived directions 0..127:
    SEPARATE = best code for each block separately
    JOINT    = best of every possible two-block code combination
  freeze both choices; test on directions 128..255
repeat after swapping the two direction halves
```

Both methods use the same alternatives, bytes, scaling calculation, and distance formula. The post-hoc test-set ceiling chooses only after seeing the test directions; it shows remaining local opportunity but is not a deployable method.

17. Joint Selection Did Not Generalize

	train half 0	train half 1
mean squared error reduction vs separate	+0.0805%	**-1.3644%**
database vectors improved	15/32	13/32
joint and separate choices differ	31/32	29/32
post-hoc test-set ceiling	16.83%	15.92%

The first row is error reduction, so larger is better; the negative value means that joint selection made the error worse.

Decision: the joint rule failed on the separate test half

Good code pairs exist after seeing the test directions, but choices learned from the other half do not predict them. The apparent opportunity comes from direction-specific error cancellation, not a stable base-trained rule.

18. Final Claim Boundary

What remains valid

Mixed-radix packing can preserve combinations of non-power-of-two level counts and prevent collisions in data deliberately having that structure.

What the natural-data evidence rejects

Neither fixed adjacent mixed-radix, optimized coordinate pairing, nor the tested base-trained two-block joint objective produces a stable new method with a better Recall–speed trade-off than the established comparisons.

Research decision

Close this line. Do not enlarge the local candidate set, change the data split or threshold, or inspect benchmark queries to rescue it. Continuing would require a genuinely new question—for example changing the distance formula or storing information shared across coordinate blocks—followed by a fresh review of the closest prior work and its costs.

References and Evidence

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